

Non-Technical Loss Investigations Management -

TGI Use Case Notes

[UC29] - Awesense Platform Use Case



Overview

This manual is intended to:

- Provide instructions to create the selected Use case in True Grid Intelligence (TGI).

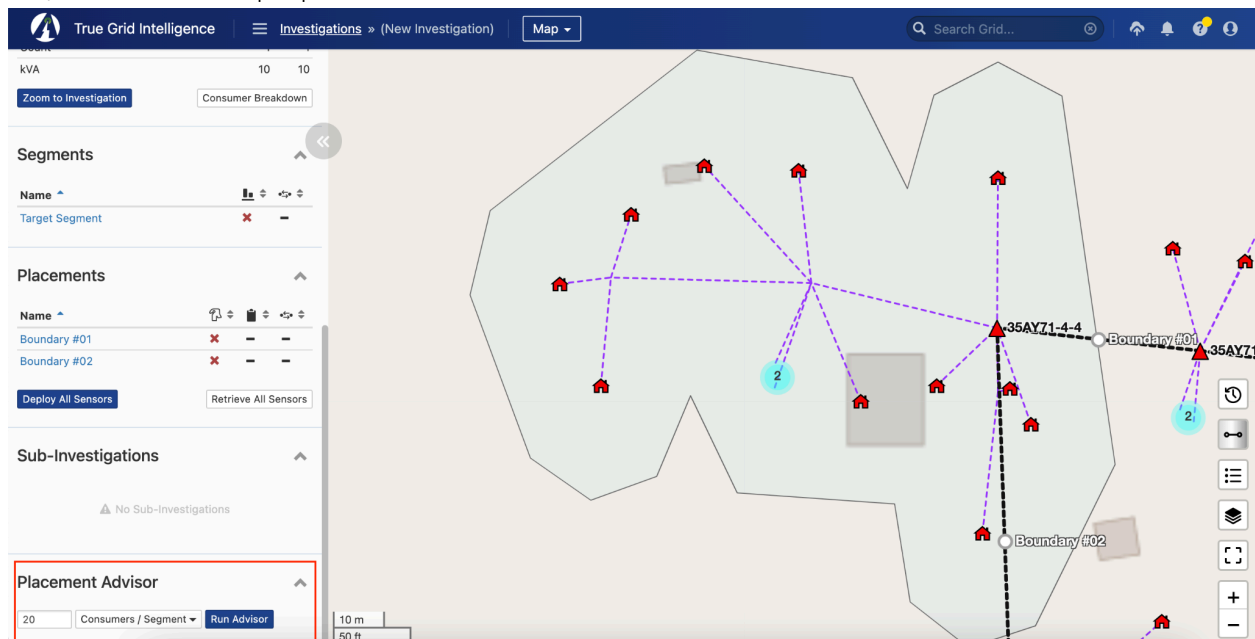
For more details about Non-Technical Loss Investigations Management and how it can be analyzed using Awesense Platform, please refer to the [UC29 - Non-Technical Loss Investigations Management.pdf](#) document.

TGI Use Case Notes

This document shows examples of using this Use Case along with a description of how to create such a Use Case on your data and screenshots directly from the TGI visualization interface.

The grid can be divided into segments using Segment Analysis - this is described in UC22. These segments can then be assigned results (the analysis factors can be defined either based on internal grid records or also on publicly available data from various trusted public sources - for example, billing irregularities, higher crime rates, customer demographics, unusual consumption patterns, type of connection, etc.) using Segment Analyzer - this is described in UC23.

This initial analysis usually divides the grid into large segments. It is necessary to create smaller segments in the newly created investigation. The "Placement Advisor", located on the bottom left, is used for this purpose.



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After selecting the rules for creating the segments into which we want to divide the grid (by the number of customers or by the number of transformers or by kVA size), we need to click on the "Run Advisor" button, which will create sub-segments.

The screenshot shows the True Grid Intelligence (TGI) web application interface. The top navigation bar includes the TGI logo, a menu icon, the text "Investigations » (New Investigation)", a "Map" dropdown, and a search bar labeled "Search Grid...". The left sidebar contains three main sections: "Segments", "Placements", and "Sub-Investigations". The "Segments" section has a table with columns "Name", "Status", and "Count". The "Placements" section has a similar table. The "Sub-Investigations" section shows a warning "No Sub-Investigations". The main area is a map showing a grid with several red house icons representing customers. Two specific locations are marked as "Placement #01" and "Placement #02" with white dots. Dashed purple lines connect these placements to various customer icons. The map also shows several grey polygonal areas representing segments. A scale bar at the bottom left indicates 10 m and 50 ft. The bottom right corner of the map has a vertical toolbar with various icons for map interaction.

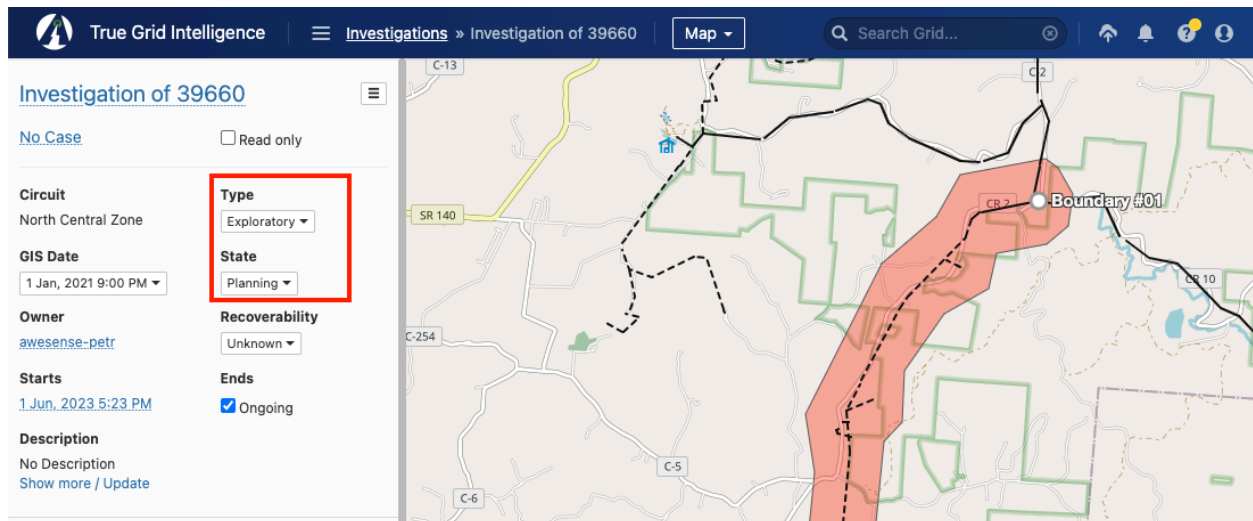
These sub-segments can be seen in the left sidebar under Segments. Placements must be deployed with measurement sensors to create energy balances.

The segment evaluation is performed in the same way as described in UC28 on page 2. For investigations, it is possible to set different types of investigations and investigation status. This then helps for easier orientation of investigation states, filtering, etc.

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For more detailed data to analyze the investigation, there is a "Report" page where it is possible to find more data about the investigation and create your own graphs.

Investigation of 39660

Overview

Case No Case	Circuit North Central Zone	Owner awesense-petr	☐ Read only
Type Exploratory	State Planning	Recoverability Unknown	GIS Date 1 Jan, 2021 9:00 PM
Starts 1 Jun, 2023 5:23 PM	Ends ☒ Ongoing	Created 22 Jun, 2023 5:23 PM	Last Modified 22 Jun, 2023 5:33 PM

Description
No Description

Energy Balances

Segment	Actual		Metered		Total Discrepancy			Days (used/total)		
	Net kWh	In kWh	Net kWh	In kWh	Out kWh	kWh				
Target Segment	146,855	146,855	0	78,808	0	78,808	68,048	3,247	46.3	21 / 53

Non-Technical Loss Investigations Management - TGI Use Case Notes

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General Note

- This description was created for TGI version 3.10.0. For future versions of TGI, the process might be slightly different.
- Access to the different parts of TGI varies depending on the permissions of the user.
- Users on the Sandbox environments currently have read-only permissions, therefore, this use case may be accessible only for viewing, not creation or editing. If you are interested in the latter, please contact Awesense.
- Some of the data for this use case may be available only on Sandbox tiers 2 & 3, but not on tier 1. If you are interested in upgrading your tier access, please contact Awesense.
- For more information please contact tools@awesense.com.